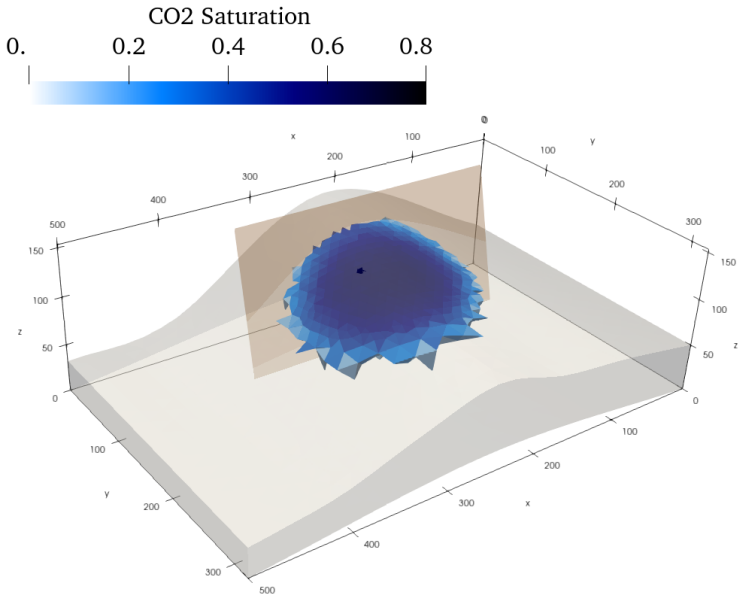
DARSim Example:

CCS with Unstructured Mesh



This example demonstrates how to use an unstructured mesh with DARSim (and coincidentally uses pEDFM as well). The mesh is generated with GMSH (see `res_mesh.msh`) and exported to a .m file format (called `res_mesh.m`). The .m mesh file is read by `unstructured_mesh.m` (note the file name inside `unstructured_mesh.m`).

To instruct DARSim to use the unstructured mesh, the simulation input file (i.e. `Sim.txt`) defines:

```
GRID_DISCRETIZATION
        UnstructuredGrid
        unstructured_mesh.m
```

This makes the definition of `DIMENSION` and `SPECGRID` keywords irrelevant. Note, though, that these keywords are still required input parameters for the simulator to run. Without them, DARSim will return an error. Therefore, these keywords are given "dummy" parameter values that are not actually used in the simulation.

This example also plots the transmissibility and pEDFM "alpha" values on mesh interfaces by using the parameter `PLOT_INTERFACES` (set to 1) in the `SimulatorSettings.txt` file. This can be helpful for visualizing the pEDFM methodology or debugging purposes.